Chapter 1, Problem 34CP

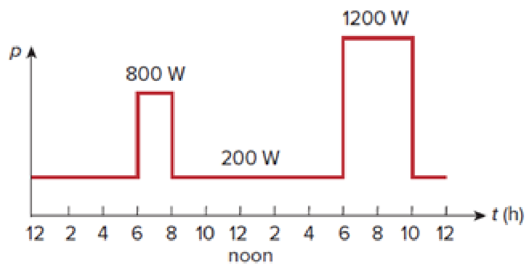
Problem

Figure 1.32 shows the power consumption of a certain household in 1 day. Calculate:

(a) the total energy consumed in kWh,

(b) the average power over the total 24 hour period.

Figure 1.32:

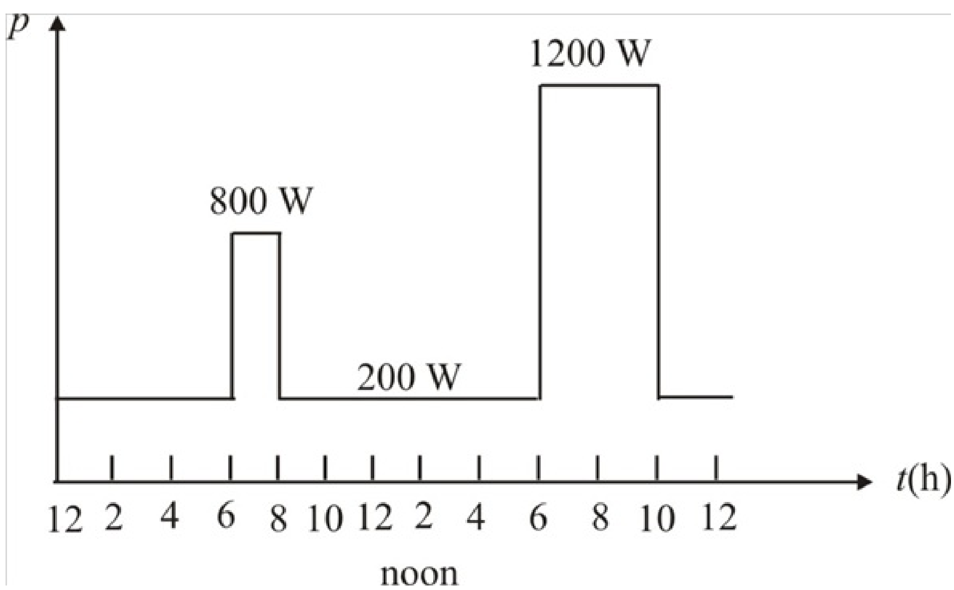


Step-by-step solution

Step 1 of 5

The power consumption of a certain household in 1 day is shown in Figure 1.

Step 2 of 5



Step 3 of 5

Figure 1

Step 4 of 5

(a)

From Figure 1, the total energy consumed is:

From 12 hours to 6 hours (6 h) @ 200W per hour $= 1200 \text{ W}$

From 6 hours to 8 hours (2 h) @ 800W per hour $= 1600 \text{ W}$

From 8 hours to 6 hours (10 h) @ 200W per hour $= 2000 \text{ W}$

From 6 hours to 10 hours (4 h) @ 1200W per hour $= 4800 \text{ W}$

From 10 hours to 12 hours (2 h) @ 200W per hour $= 400 \text{ W}$

The total energy consumed for 24 hours $= 10000 \text{ W}$

Therefore the total energy consumed for 24 hours in kWh is $\boxed{10 \text{ kWh}}$.

Step 5 of 5

(b)

The total energy consumed for 24 hours $= 10000 \text{ W}$

Calculate the average power per hour over 24 hour period.

$$\begin{aligned} \text{average power per hour} &= \frac{10000 \text{ W}}{24} \\ &= 416.66 \text{ W} \end{aligned}$$

Therefore, the average power per hour over 24 hour period is $\boxed{416.66 \text{ W}}$.